Homework 1

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Problem 1 part 1

A matrix A is invertible if $det(A) \neq 0$. To show that a monomial matrix is invertible, the same property must apply. To prove that a monomial matrix is invertible, we will take the determinient of the base case, a 2×2 matrix, and the inductive case, a $k \times k$ matrix with $k > 2$, are both non zero.

Say $A \in \Re^{2\times 2}$ is a monomial matrix. The deterinant of A is $ad - cb$, where the variables come from $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. Since the matrix is monomial, either ac or $-bd$ evaluates zero, as there can only be one non-zero value per column and row in the matrix. Therefore, $det(A) \neq 0$.

For the inductive step, where $A \in \Re^{k\times k}$, the determinient can be calculated in the same way as a diagonal matrix, the product of all non-zero entries of the matrix. A monomial matrix requires that A only has one non-zero entry per row and column, the same as a diagonal matrix. To convert a monomial matrix to a diagonal matrix, the rows of the matrix need to be rearranged where $A_{ii} \neq 0$. In a 2×2 monomial matrix, re arranging the rows affects the determinient as follows, $bd - ac$. The tranformation is just multiplying the the determinant by -1. If we apply another rarranging of the rows, the determinant goes back to its original formulation, essentially multiplying the determinant by -1^2. Applying this to a general case, if rearranging l rows of a matrix results in the deterinant being multiplied by $(-1)^l$, then the determinant of a monomial matrix is, after rearranging the rows to make it diagonal, $(-1)^l \prod_{i=1}^{n}(A_{ii})$, where l is the number of row swaps being made. Therefore, for a $k \times k$ monomial matrix, the determinant is non-zero. By proof of induction, a monomial matrix is invertible.

Problem 1 part 2

Given A is skew symmetric, $A^{\mathbf{T}} = -A$. In quadratic form, $x^{\mathbf{T}}Ax = 0$. By transposing the entire expression, $(x^TAx = 0)^T = xA^Tx^T = x(-A)x^T$. The resulting expression is $x^TAx = -xAx^T$, Adding the right hand side to the left gives us $2x^TAx = 0$, rewrittting the equation as $x^TAx = 0$ by dividing by 2. Therefore $x^TAx = 0$.

Problem 2 part 1

$P_2E_1 = \left(\begin{smallmatrix} 1 & 0 & 0 & 0 \\ l_{41} & 0 & 0 & 1 \\ l_{21} & 1 & 0 & 0 \\ l_{31} & 0 & 1 & 0 \end{smallmatrix}\right)$. $P_2^{-1} = P_2$ since the gaussian elimination only requires the swapping of rows 4, 2 and 3 on the identity matrix. $P_2E_1P_2^{-1} = P_2E_1P_2 = \tilde{E_1}$.

$$\tilde{E_1} = \left(\begin{smallmatrix} 1 & 0 & 0 & 0 \\ l_{41} & 0 & 1 & 0 \\ l_{21} & 0 & 0 & 1 \\ l_{31} & 1 & 0 & 0 \end{smallmatrix}\right)$$

.

Problem 2 part 2

$\tilde{E_k}$ is different from E_k by the fact that $\tilde{E_k}$ has both its rows $i,\ j$ and columns $i,\ j$ permuted. As seen by the first part of this problem, when E_k is multiplied by the permutation matrix from the left, only the rows of the E_k are permuted. If it is multiplied from its right, its columns are permutted. Since $\tilde{E_k}$ needs to equal E_k with its rows permuted, $\tilde{E_k}$ needs to have both its rows and columns permutted to satisfy the equation.

Problem 3 part 1

In order to get $A = LU$, the following row operations need to be executed.

1. $R_2 = R_2 - (2R_1)$
2. $R_3 = R_3 - (\frac{3}{2}R_2)$

This results in $L = \left(\begin{smallmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & \frac{3}{2} & 1 \end{smallmatrix}\right)$ and $U = \left(\begin{smallmatrix} 6 & 10 & 0 \\ 0 & 6 & 4 \\ 0 & 0 & 6 \end{smallmatrix}\right)$.

To solve for $Ly = b$, we use forward substitution, where $y_1 = 4,\ y_2 = 10 - 8 = 2,\ y_3 = -3 - 3/2\ *\ (2) = -3 - 3 = -6$. Then, to solve $Ux = y$, we use backwards substitution, where $x_3 = -6/6 = -1,\ x_2 = (4+2)/6 = 1,\ x_1 = (4 - 10)/6 = -1$.

$$x = \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}$$

Problem 3 part 2

$$A = \begin{pmatrix} 1 & -5 & 1 \\ 10 & 0 & 20 \\ 5 & 0 & -1 \end{pmatrix},\ b = \begin{pmatrix} 7 \\ 6 \\ 4 \end{pmatrix}$$

In order to solve this, we need to create a permutation matrix P by swapping the first row with the second row of A.

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix},\ \tilde{A} = \begin{pmatrix} 10 & 0 & 20 \\ 1 & -5 & 1 \\ 5 & 0 & -1 \end{pmatrix},\ \tilde{b} = \begin{pmatrix} 6 \\ 7 \\ 4 \end{pmatrix}$$

Now we can use row operations to obtain $PA = LU$. The row operations are as follows.

1. $R_2 = R_2 - (\frac{1}{10}R_1)$
2. $R_3 = R_3 - (\frac{1}{2}R_1)$

$$L = \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{10} & 1 & 0 \\ \frac{1}{2} & 0 & 1 \end{pmatrix},\ U = \begin{pmatrix} 10 & 0 & 20 \\ 0 & -5 & -1 \\ 0 & 0 & -11 \end{pmatrix}.$$

With both triangular matrices, we can solve $Ly = b$, where $y_1 = 6,\ y_2 = 7 - \frac{6}{10} = \frac{32}{5},\ y_3 = 4 - \frac{32}{5} * \frac{1}{2} = 1$. Solving $Ux = y$ gives $x_3 = \frac{-1}{11},\ x_2 = \frac{\frac{32}{5} - \frac{1}{11}}{5} = \frac{-347}{275},\ x_1 = \frac{6+20*\frac{1}{11}}{10} = \frac{86}{110}$.

$$x = \begin{pmatrix} \frac{86}{110} \\ \\ \frac{-347}{275} \\ \\ \frac{-1}{11} \end{pmatrix}$$

Problem 4

The following graphs plot $(x_i,\ u_i)$ for each $n = 1,\ 2,\ \ldots,\ 8$.

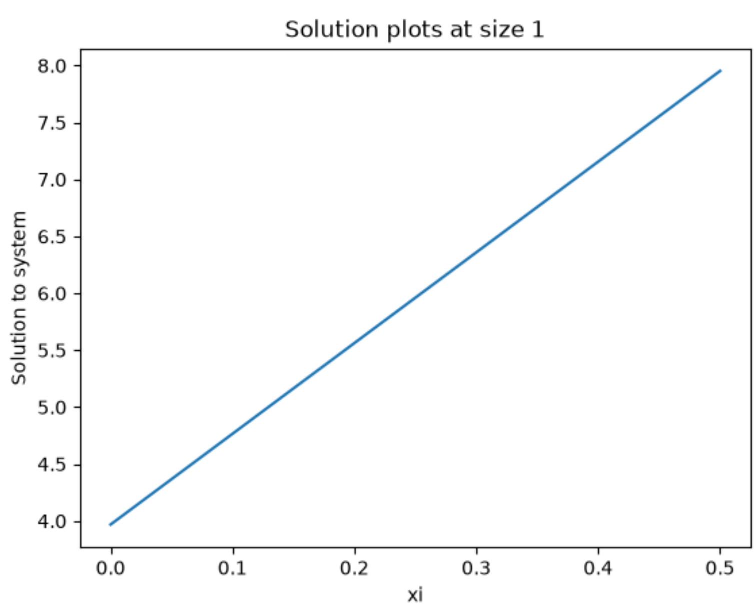

Figure 1: the solution to the system u_i with $n = 1$ and x_i

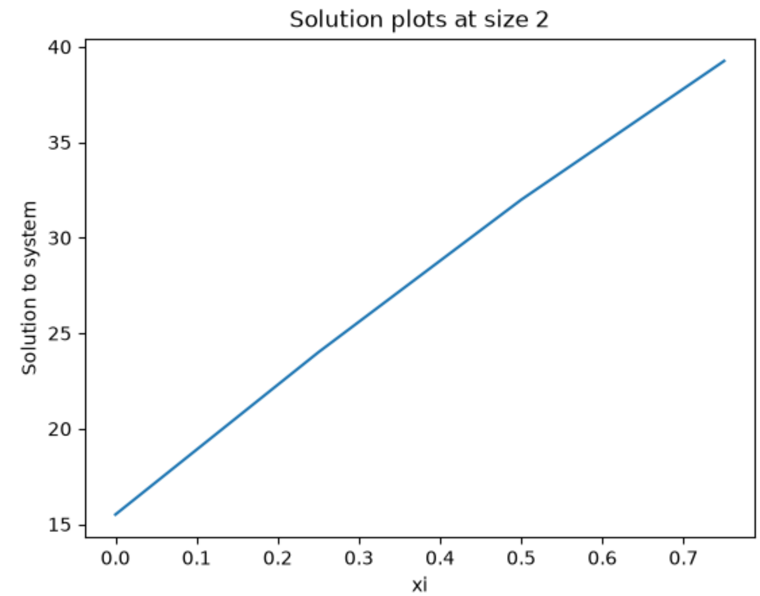

Figure 2: the solution to the system u_i with $n = 2$ and x_i

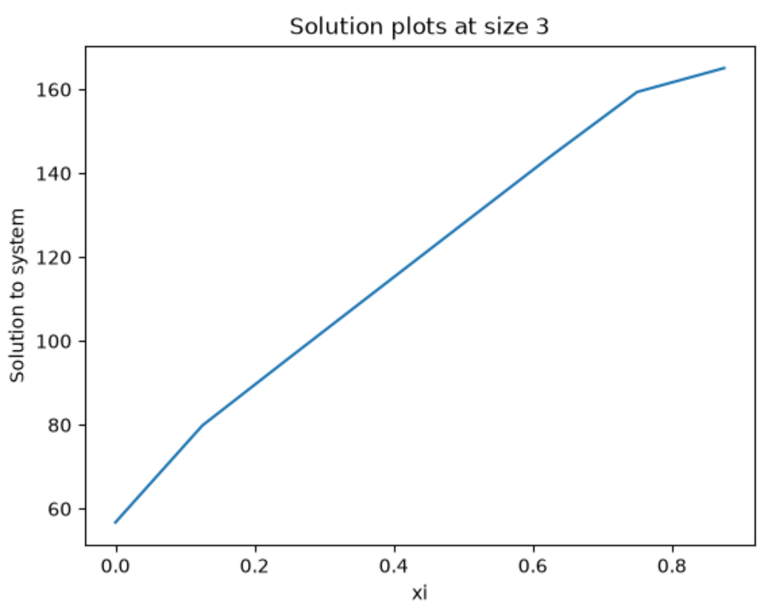

Figure 3: the solution to the system u_i with $n = 3$ and x_i

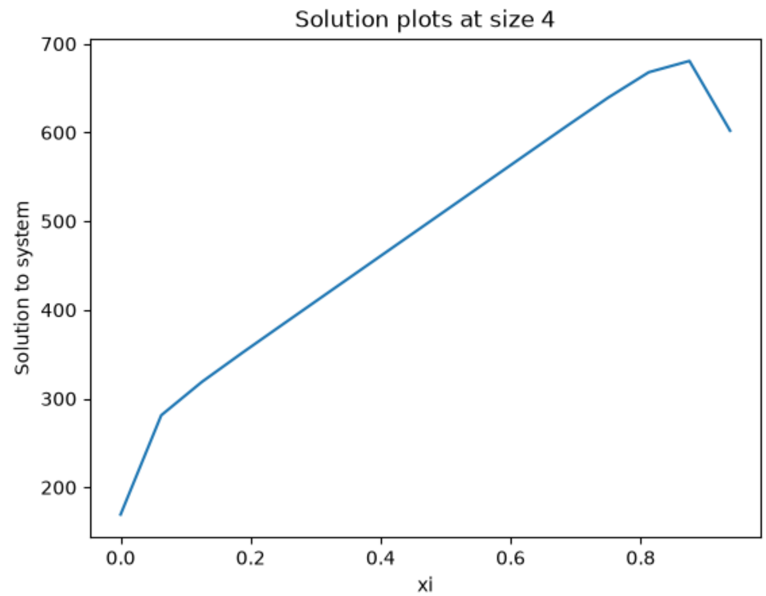

Figure 4: the solution to the system u_i with $n = 4$ and x_i

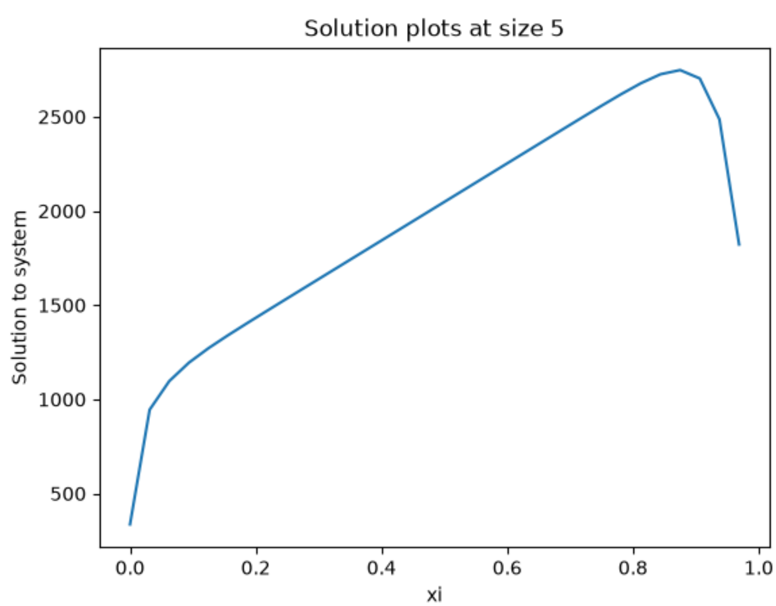

Figure 5: the solution to the system u_i with $n = 5$ and x_i

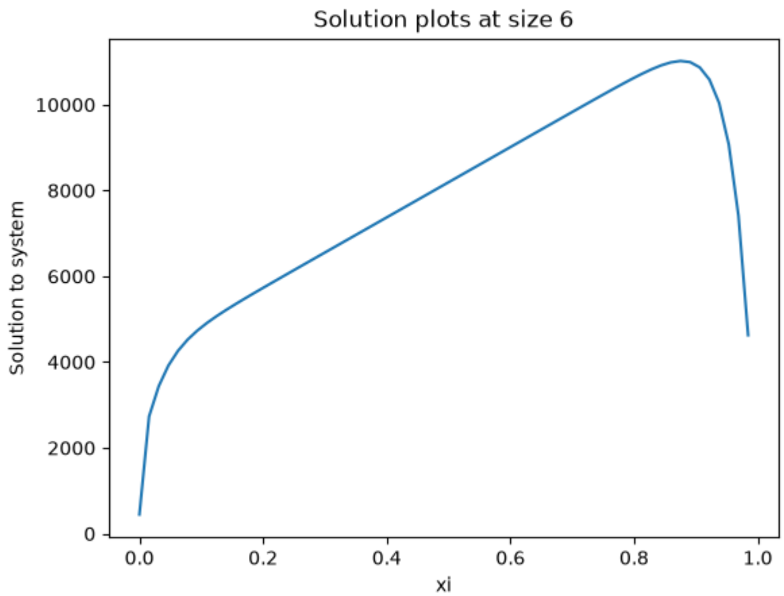

Figure 6: the solution to the system u_i with $n = 6$ and x_i

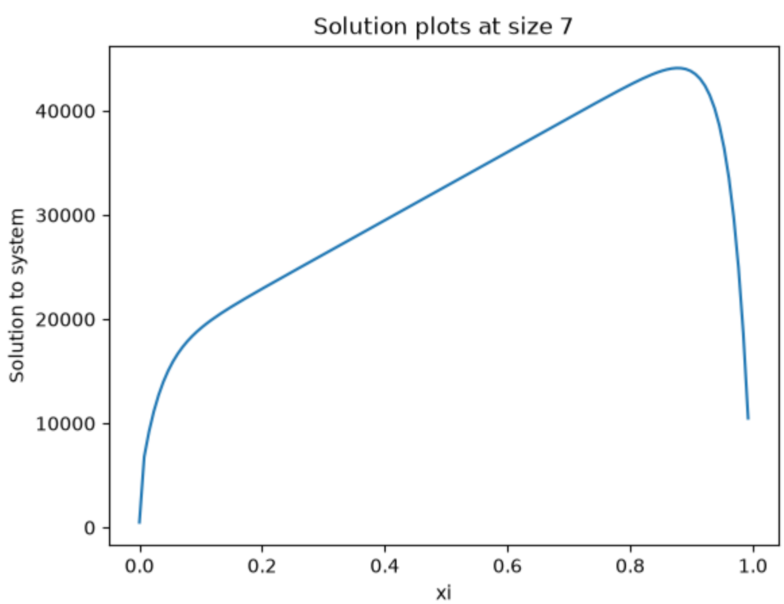

Figure 7: the solution to the system u_i with $n = 7$ and x_i

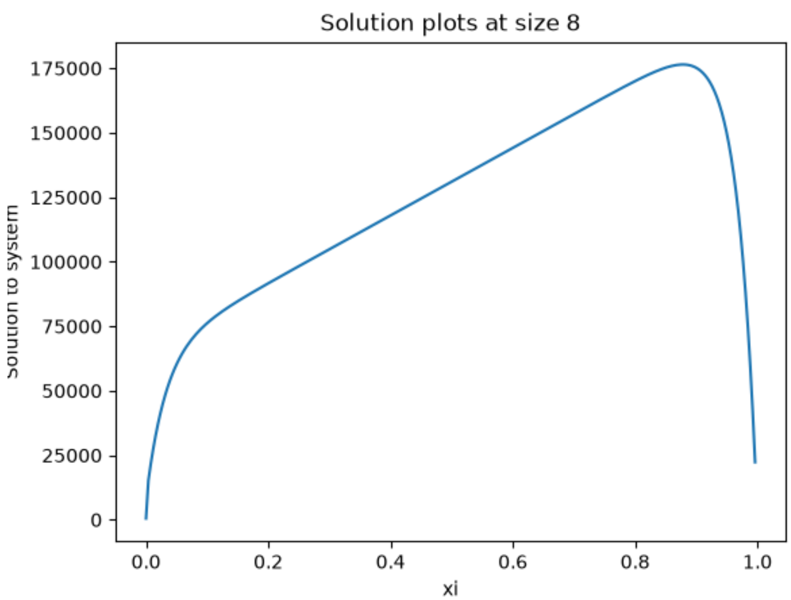

Figure 8: the solution to the system u_i with $n = 8$ and x_i

Problem 5

The table below shows the solution to the problem defined in the document.

Table 1: Solutions for matrix for every n

n	Solutions
4	$[10.0019,\ 0.0551,\ 0.0177,\ 0.6021]$
5	$[14.6843,\ 8.743 \times 10^{-3},\ 5.212 \times 10^{-4},\ 1.570 \times 10^{-5},\ 1.0185]$
6	$[27.3270,\ 2.438 \times 10^{-3},\ 2.620 \times 10^{-5},\ 2.328 \times 10^{-7},\ -1.177 \times 10^{-9},\ 1.1146]$
7	$[54.7108,\ 6.207 \times 10^{-4},\ 1.110 \times 10^{-6},\ -8.239 \times 10^{-9},\ 0,\ 0,\ 1.1295]$
8	$[111.3813,\ 1.266 \times 10^{-4},\ 3.279 \times 10^{-8},\ 1.191 \times 10^{-11},\ 1.316 \times 10^{-13},\ 0,\ 0,\ 1.1220]$
9	$[226.7041,\ 2.103 \times 10^{-5},\ 7.008 \times 10^{-10},\ 0,\ 2.768 \times 10^{-12},\ -1.723 \times 10^{-13},\ 1.513 \times 10^{-14},\ 1.881 \times 10^{-15},\ 1.1105]$
10	$[459.8511,\ 2.950 \times 10^{-6},\ 4.459 \times 10^{-10},\ 0,\ -7.022 \times 10^{-11},\ 0,\ -3.755 \times 10^{-13},\ 0,\ 2.427 \times 10^{-15},\ 1.0999]$
11	$[929.9248,\ 3.597 \times 10^{-7},\ -2.130 \times 10^{-9},\ 0,\ 0,\ -4.309 \times 10^{-11},\ -2.916 \times 10^{-12},\ 0,\ -1.809 \times 10^{-14},\ 0,\ 1.0909]$
12	$[1876.3377,\ 5.979 \times 10^{-8},\ 3.722 \times 10^{-8},\ 0,\ -9.063 \times 10^{-8},\ -7.858 \times 10^{-10},\ -7.323 \times 10^{-11},\ -3.780 \times 10^{-12},\ 0,\ -2.371 \times 10^{-14},\ 0,\ 1.0$
13	$[3779.9242,\ -9.553 \times 10^{-7},\ 0,\ 0,\ 0,\ 3.410 \times 10^{-8},\ -1.838 \times 10^{-9},\ 0,\ 1.417 \times 10^{-11},\ -5.747 \times 10^{-13},\ 0,\ 0,\ 1.0769]$
14	$[7605.8574,\ 1.527 \times 10^{-5},\ 0,\ 0,\ 0,\ -5.205 \times 10^{-7},\ -2.613 \times 10^{-8},\ 2.506 \times 10^{-9},\ 0,\ -1.118 \times 10^{-11},\ -4.824 \times 10^{-13},\ 0,\ 2.961 \times 10^{-15},$
15	$[15290.7327,\ 0,\ 0,\ 0,\ 0,\ 2.208 \times 10^{-5},\ -7.359 \times 10^{-7},\ 0,\ -2.509 \times 10^{-9},\ 0,\ 0,\ -6.113 \times 10^{-13},\ 0,\ 2.388 \times 10^{-15},\ 1.0667]$
16	$[30719.0000,\ 0,\ 0,\ 0,\ 0,\ 0,\ 0,\ 1.055 \times 10^{-6},\ 0,\ 2.878 \times 10^{-9},\ -2.352 \times 10^{-10},\ 0,\ -1.887 \times 10^{-12},\ 4.626 \times 10^{-14},\ 0,\ 1.0625]$
17	$[61680.0821,\ 0.2500,\ -0.1521,\ 0,\ 0,\ 0,\ -5.123 \times 10^{-4},\ 0,\ 0,\ 8.848 \times 10^{-8},\ -4.309 \times 10^{-9},\ 0,\ -1.456 \times 10^{-11},\ 0,\ 0,\ 0,\ 1.0588]$
18	$[123782.3843,\ 4.0003,\ 0,\ 0,\ 0,\ 0,\ -0.0197,\ 8.601 \times 10^{-4},\ -6.266 \times 10^{-5},\ 2.799 \times 10^{-6},\ 0,\ 9.416 \times 10^{-9},\ -3.534 \times 10^{-10},\ 0,\ 0,\ 0,\ 0,\ 1.$
19	$[248525.3548,\ -64.0024,\ 36.9062,\ 0,\ 12.6922,\ -10.9437,\ -0.8575,\ 0.0264,\ -1.722 \times 10^{-3},\ 7.359 \times 10^{-5},\ 3.949 \times 10^{-6},\ 0,\ 0,\ -5.827 \times 1$
20	$[491599.6122,\ 0,\ 0,\ 0,\ 203.4072,\ 0,\ 50.2885,\ 0,\ 0,\ 0,\ 0,\ 5.865 \times 10^{-6},\ -2.537 \times 10^{-7},\ 0,\ 0,\ 0,\ -1.685 \times 10^{-12},\ 0,\ 0,\ 1.0500]$

Discussion

The solutions shown above are obviously not correct. All of the solutions should be all 1s, but instead, there are zeros, really small decimals, and large numbers. This is due to the round of errrors introducted by floating point computations. A 64 bit floating point integer has a fixed amount of bits to represent the mantissa of a decimal value. Because of the limited number bits to represent a decimal, there is some amount of error created when representing a small decimal number. When using these approximated numbers in computations, espescially in numerically unstable algorithms like gaussian elimination, the error of said approximation gets propagated throughout the operations, resulting in numbers that are incorrect.